

Chapter 3 _ Inside the mind of the Agentic AI

An AI agent functions more like a skilled digital employee who can think, adapt, and handle complex situations independently. it can manage end-to-end processes, make decisions, and adjust its approach based on the context.

Specialty:

- **24/7 operation:** different timezone, everywhere
- **Infinite scalability:** easy to deploy, ability to adapt to new situation, learn new skills
- **Universal Applicability:** it can function effectively across different industries and business functions. it can be applied broadly across sectors. The SPAR remain consistent whether they are analyzing the banking system...
- **Collaboration:** it can collaborate with human and other agents. each have its own specific role

Limitation

- **The simulation of intelligence:** they dont understand the world in the way that human do. they are simply making the predictions based on past patterns about what comes next.
- **The data quality dilemma:** it depend on the quality of the data. the llm would react to the prompt based on what it was trained on whether it is correct or not. and they do not know if there are errors in their output.
- **The common-sense gap:** the most striking challenge is that they do not have the common- sense reasoning. they can produce incredible result, but they often miss the real-world constraints that any human would notice instantly.
- **The creativity Conundrum:** they struggle with true creativity and innovation. they produce mixed existing patterns, but rarely generate the original ideas. It wont produce any masterpiece that could shape the industry.
- **The hallucination problem:** it can generate false information with high confidence. the human oversight is crucial here
- **The ethic and judgement gap:** while they can be programmed with rules and guideline, they don't understand the deeper moral implications of their decisions or the complex human factors that often need to be considered.

Solution

- Creating hybrid systems where AI agents handle the data-intensive tasks while humans provide oversight and judgement.
- Establishing strong validation methods to identify possible hallucination problems or mistakes, such as having various agents assess the output of their peers.
- Applying strategies to enhance the quality of the agent input data

- Maintaining human involvement in decisions that require ethical reasoning or complex contextual understanding
- Developing transparency so that humans can understand and evaluate how agents make decisions and take actions

Multi-Agent Systems

- Complexity management
- specialized agent perform better at specific tasks than generalist systems.
- super resilience: if one agent fail, or need the maintenance, the other can continue or often adapt to cover the gaps.

Framework for the MAS

1. The one agent, one principal tool

One-agent, One-tool: assign one agent a specific tool, each agent can truly master its domain, doing one thing but doing it right!

One coordinator: that control the flow and translate everything between all the agent to ensure smooth workflow

2. **The hierarchical Model:** The agent are organized in layers, with higher-level agents coordinating the activities of those below them. it is in between the centralized and decentralized approach
3. **Centralized control:** the central agent maintain a global view of the system's state and goals, making high-level decisions that other agents execute. while it can provide the clear and strong consistency, it can create the single point of failure.
4. **Decentralized control:** all agents are peers, coordinating through direct communication without any central authority. This approach offer the scalability and resilience but required the sophisticated coordination protocols to ensure the agent work together effectively.

To make it work

The success of MAS depend on the several critical factors:

- **Clear communication protocols:** for sharing information and coordinating systems, designing robust communication protocols is a must
 - Add the Translator Agent whose job is to ensure the smooth communication between different parts of the systems.
- **Coordination mechanisms:** Coordinating strategies (scheduling the rules: Agent A goes first, then Agent B) or a more complex algorithm (like multi agent planning or negotiation protocols)
- **Robustness and Error recovery:** The backup coordinator that took over when the primary failed during the live demo. if one agent start to behave erratically, the other agent can flag or ignore the unreliable data or flag it to human reviewer

- **Emergent behavior in MAS:** the agents can sometimes exhibit emergent behaviors like the complex strategies or solutions that were not explicitly programmed. Collective intelligent is when the model collaborate or even just coexist. The agent can engage in complex social collaboration, negotiating, persuading, and coordinating actions in a group setting.

Stochasticity

The Model does not simply follow the rule: they generate the response based on the patterns learned from a massive amount of training data.

Stochasticity = refers to the randomness inherent in the way that the model generates the response.

Benefit of the stochasticity

- Natural interaction: customer service...
- Problem solving: generating different approaches lead to a better solution
- Learning and flexibility: trying different approaches and learning from various experiences

The issues with stochasticity

- Consistency
- Precision: numerical accuracy, specific format

Solution to the stochasticity

- Temperature control (Creativity Control)